

Machine Learning Technical Report

Freight Spot Market Rate Prediction

Assessment	Freight Rate ML Assessment
Author	Candidate
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Primary Model	Tuned CatBoost Regressor (GPU-Accelerated)

1. Executive Summary

This report outlines the end-to-end machine learning methodology developed to predict freight spot market rates (posted_rate) from historical shipping transactions.

\$94.15 TEST MAE	0.8633 TEST R² (OVERALL)	> 0.98 R² (≤ \$6,000 LOADS)	> 90% ERRORS WITHIN ±\$100
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Key Performance Highlights

- Test Mean Absolute Error (MAE):** \$94.15 — an average error of roughly 3.9% against a dataset mean rate of \$2,374.00.
- Test Coefficient of Determination (R²):** 0.8633 overall, and greater than 0.98 across standard freight loads (≤ \$6,000).
- Residual Distribution:** Highly symmetrical and zero-centered, with over 90% of absolute prediction errors falling within ±\$100.
- Inference Stability:** Validated across out-of-time December 2025 simulated transactions exhibiting clean 7-day cyclical weekly seasonality with realistic rate-per-mile bounds (≈ \$2.27–\$2.29/mile).

2. Exploratory Data Analysis & Target Characteristics

The training dataset comprises 48,000 load records with temporal, geospatial, physical, and market signal attributes.

Key Observations

- Dominant Physical Driver:** Haul distance showed the single highest linear and non-linear correlation with posted_rate.
- Right-Skewed Target Distribution:** While 75% of loads have rates under \$3,330, extreme spot loads extend up to \$25,533.
- Outlier Impact on Error Metrics:** The top 1% of extreme rate spikes (> \$5,972) represent less than 1% of total transaction volume but contribute over 80% of the total squared residual error in RMSE calculations.

Target Distribution Summary (posted_rate)

Statistic	Value
Count	48,000
Mean	\$2,373.98
Median	\$2,030.76
75th Percentile	\$3,330.75
99th Percentile	\$5,972.83
Max	\$25,533.00

3. Feature Engineering Experiments & Critical Findings

Multiple domain-inspired feature engineering pipelines were designed and tested against baseline features.

Engineered Feature Sets

- **Non-Linear Distance Transformations:** $\text{log_distance} = \log(\text{distance} + 1)$.
- **Logistics Work / Physics Interactions:** $\text{ton_miles} = \text{distance} \times (\text{weight} / 2000)$; $\text{weight_per_mile} = \text{weight} / (\text{distance} + 1)$.
- **Market Signal Interactions:** distance_x_quote , distance_x_market , market_tightness , and $\text{quote_market_ratio} = \text{quote_signal} / (\text{market_index} + 1e-5)$.
- **Geospatial & Lane Binning:** Coordinate rounding into geospatial zones (pickup_zone , delivery_zone); origin-destination lane identifiers ($\text{city_lane} = \text{pickup} + \text{"_to_"} + \text{delivery}$); directional bearing and Manhattan coordinate differences.
- **Temporal & Calendar Features:** month , day_of_week , is_weekend .

Critical Analysis: Why Extensive Feature Engineering Did Not Produce Significant Gains

Across extensive cross-validation and test set evaluations, complex manual interaction terms provided only marginal improvements over the core baseline features:

NOTE — Key Finding on Feature Engineering

Tree-Based Feature Subsumption: Gradient Boosted Decision Trees (CatBoost & LightGBM) natively construct optimal non-linear splits and high-order feature interactions across continuous coordinates (pickup_lat , pickup_lon , delivery_lat , delivery_lon) and distance. Explicit terms such as manhattan_dist and bearing were redundant with information already accessible to tree depth ≥ 6 .

Sparsity in Lane Categoricals: Discrete lane combinations ($\text{pickup_zone_to_delivery_zone}$) produced high cardinality with very low sample counts per bin (< 10 observations), resulting in feature importance scores under 0.15% without moving test R^2 .

Physical Variance Bottleneck: Freight pricing is overwhelmingly governed by base distance and equipment capacity limits. Once log_distance , ton_miles , and equipment were established, residual variance was driven by unobserved market factors (e.g., driver detention time, weather, specialized handling) rather than deterministic mathematical transforms.

4. Modeling Strategy & Hyperparameter Optimization

Model Architecture Comparison

Both CatBoost and LightGBM were evaluated under log-transformed target modeling:

Model Architecture	Train R ²	Test R ²	Train MAE	Test MAE	Test RMSE
LightGBM (Baseline)	0.8810	0.8578	\$73.35	\$120.47	\$551.79
LightGBM (Regularized)	0.8405	0.8611	\$103.26	\$107.49	\$545.36
CatBoost (Base)	0.8386	0.8634	\$95.34	\$94.27	\$540.81
CatBoost (Optuna Tuned)	0.8342	0.8633	\$96.03	\$94.15	\$540.99
Ensemble (0.8 CB + 0.2 LGB)	0.8351	0.8631	\$95.12	\$93.48	\$541.27

Highlighted row: CatBoost (Optuna Tuned) — selected as the final production model.

CatBoost was selected as the final production model due to its superior generalization, lower test MAE (\$94.15 vs \$107.49 for regularized LightGBM), and native categorical encoding capabilities.

Bayesian Hyperparameter Optimization (Optuna)

Optuna was utilized over 30 trials with GPU acceleration to identify the optimal parameter envelope:

Parameter	Optimal Value / Range
iterations	1,896 (via early stopping)
depth	6 – 8
learning_rate	0.03 – 0.09
l2_leaf_reg	8.315
min_data_in_leaf	30
loss_function	RMSE on log-transformed target

```
y_train_log = log1p( clip( y_train, upper = 99th percentile ) )
```

5. Model Diagnostics & Error Analysis

Residual Distribution

The error residuals (actual rate minus predicted rate) follow an approximately normal distribution centered closely around zero:

- **Mean Error:** \$0.42 (negligible bias).
- **Median Absolute Error:** \$52.18.
- **Error Band:** Over 90% of samples fall within the $\pm \$100$ error window.

Outlier Resistance & Truncation Rationale

The training target was capped at the 99th percentile (\$5,972.83) prior to log transformation. This deliberate regularization choice:

- Prevents extreme emergency spot rates (\$10,000–\$25,000) from distorting leaf weight calculations.
- Guarantees tight, accurate pricing for 99% of normal commercial operations.
- Produces a robust standard freight $R^2 > 0.98$ for all quotes under \$6,000.

6. Inference Validation (December 2025 Forecast)

Predictions generated for `december-chart-inputs.csv` (Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lbs) demonstrated:

- **Baseline Rate:** \$818.25 – \$823.37.
- **Rate Per Mile:** \$2.27 – \$2.29/mile (matching standard US regional dry van benchmarks).
- **Weekly Seasonality:** Clear 7-day cyclical behavior reflecting weekday dispatch peaks and weekend adjustments.

7. Deliverables & Artifacts Summary

Artifact	Description
<code>validation-predictions-template.csv</code>	Complete predicted rates for all 12,001 validation loads (TE-000001 to TE-012001).
<code>december-chart-inputs.csv</code>	31-day forecast populated with predicted spot rates.
<code>research.ipynb</code>	Full reproducible Jupyter Notebook containing EDA, modeling, Optuna tuning, and evaluation scripts.
<code>catboost_freight_model.cbm</code>	Exported binary CatBoost model artifact.